

Budget

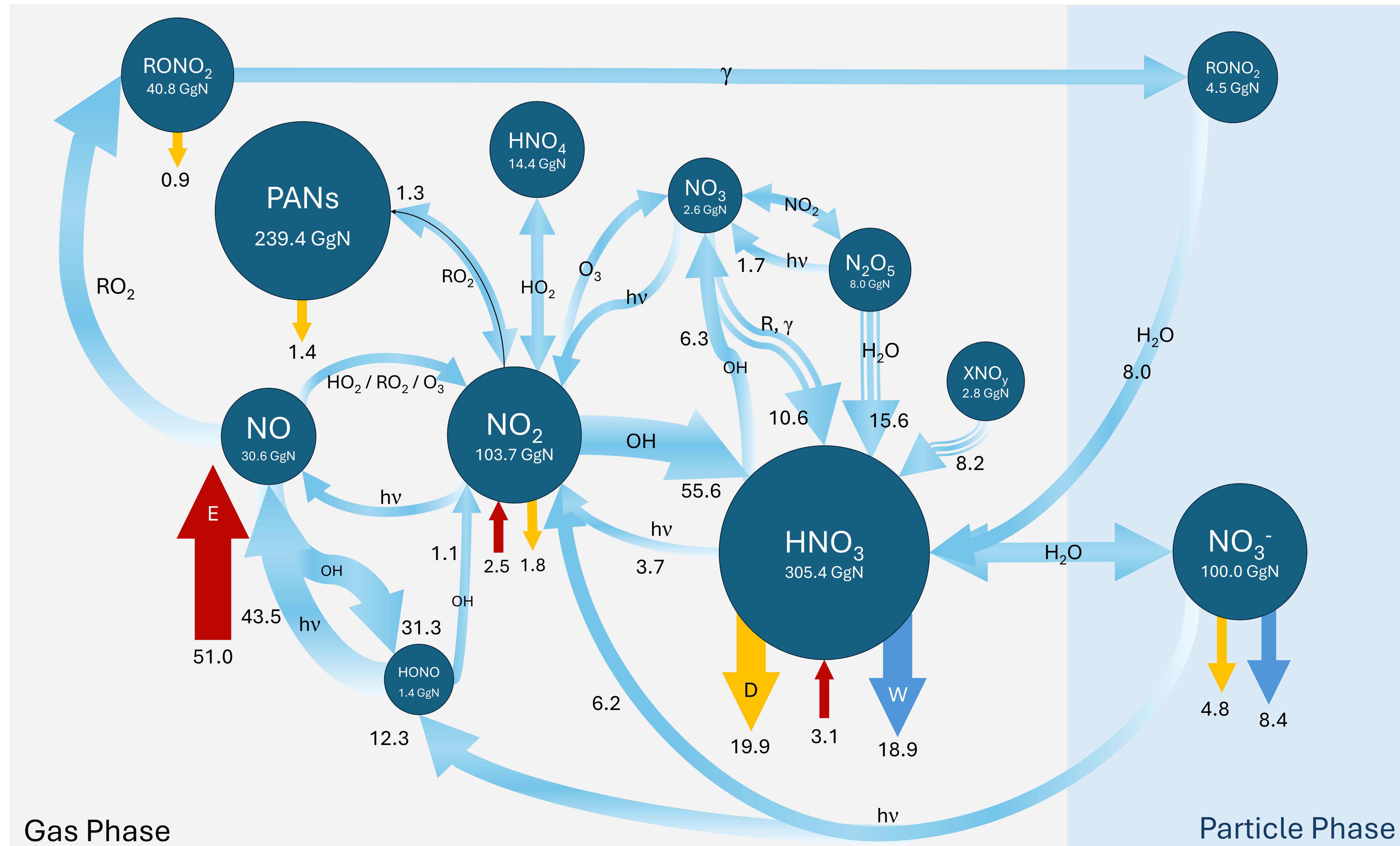


Figure 1: Modeled NO_y budget showing key species (blue circles) in GgN for 2016. Physical fluxes (emissions, wet deposition, and dry deposition) exceeding 0.5 TgN/yr are represented by vertical arrows. Chemical fluxes exceeding 1 TgN/yr are represented by curved light blue arrows.

Seasonality

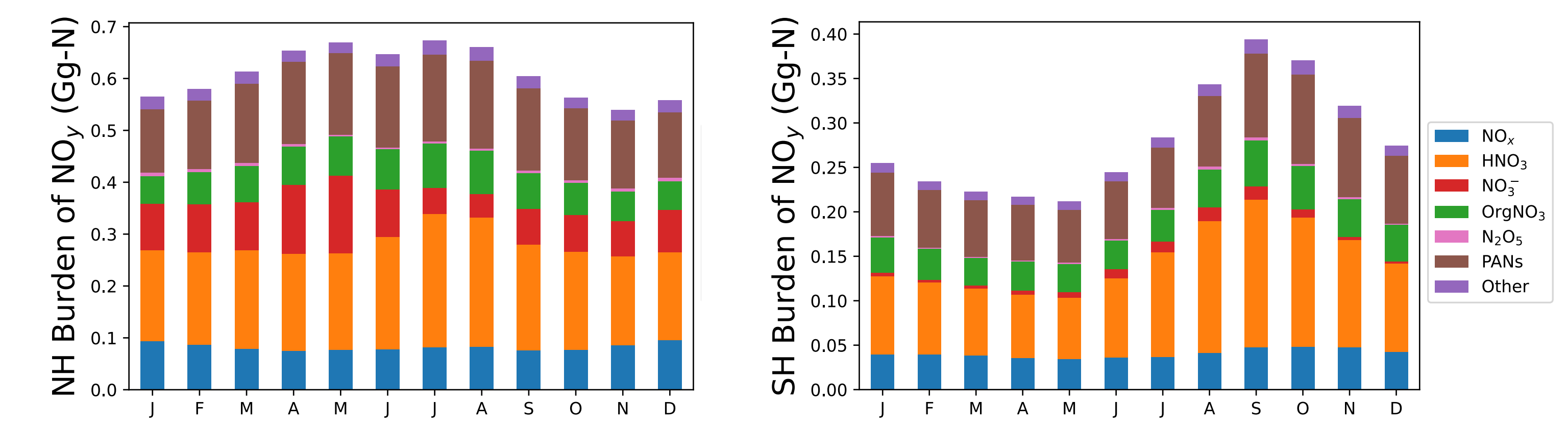


Figure 4: Modeled seasonal changes in the global burden of NO_y species in 2016.

Observations

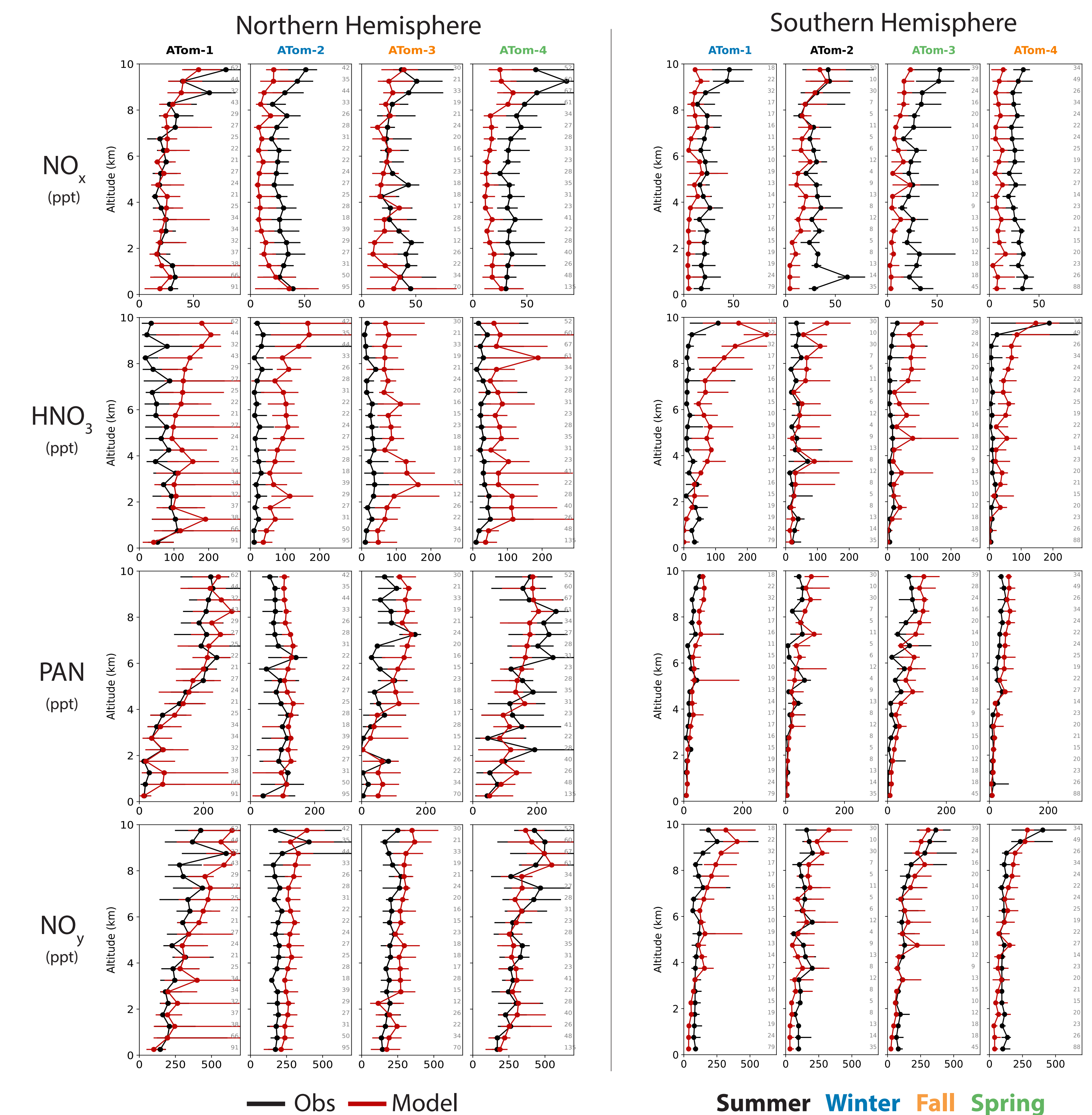


Figure 5: Modeled and observed (median) vertical profiles of NO_y species during ATom flights. Campaign label colors indicate hemispheric seasons across the mission.

- GEOS-Chem underestimates NO_x throughout the troposphere, across all seasons except boreal summer.
- The model slightly overestimates NO_y largely due to the overestimate of HNO_3 , indicating either excess production or inadequate removal of NO_y .
- This high model bias in NO_y taken with the low model bias in NO_x leads to an overall low bias of $\text{NO}_x : \text{NO}_y$ with only boreal summer being well-represented.

Lifetime

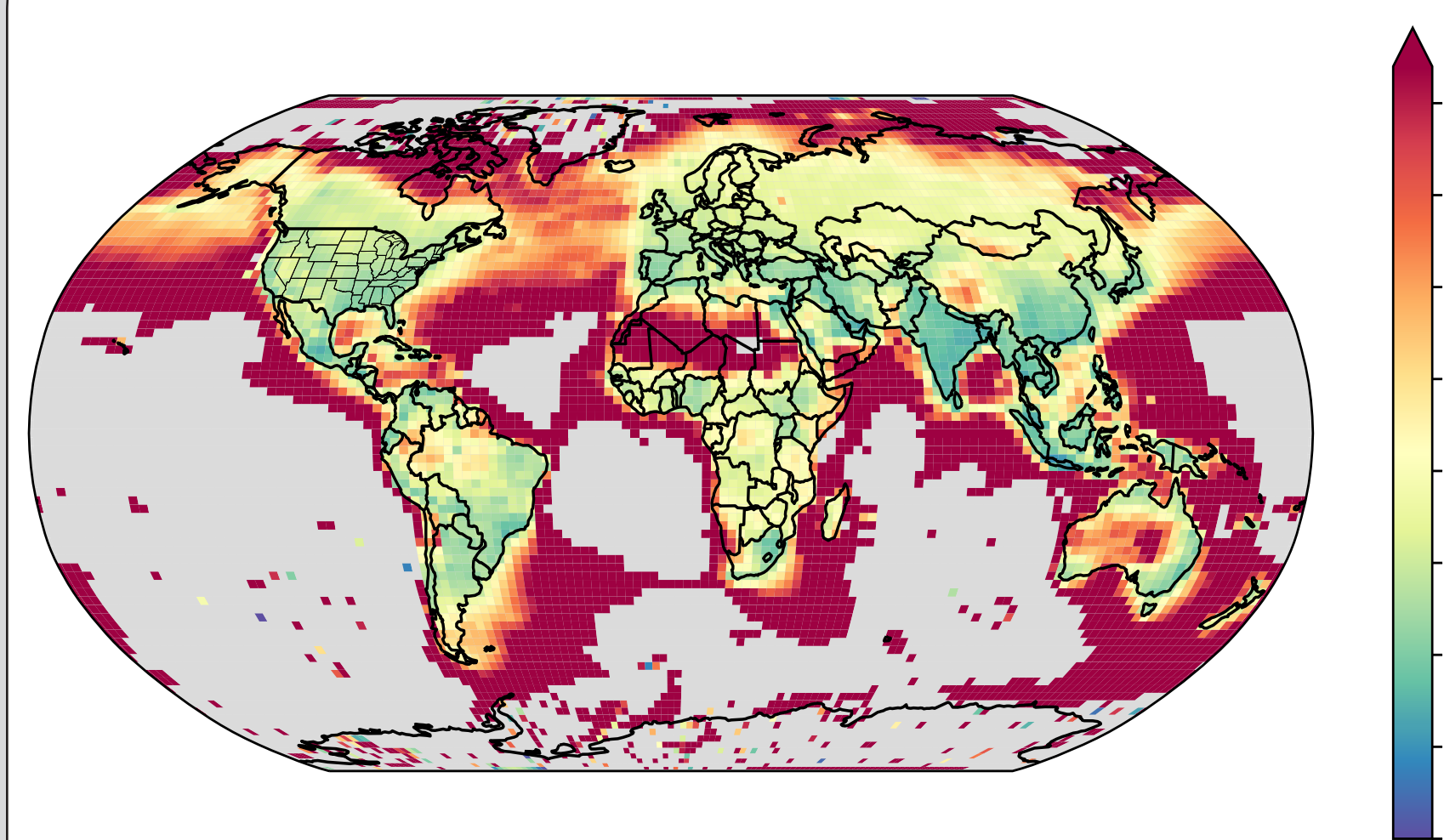


Figure 2: Modeled NO_x column chemical lifetime (in days). Lifetimes > 5 days are masked out.

- The mean global lifetime of NO_x is 21 hours. NO_x lifetime is governed by the rate of oxidation of NO_2 to HNO_3 by OH and to PANs by RO_2 near the surface where NO_x burdens are highest.

$\text{NO}_x : \text{NO}_y$

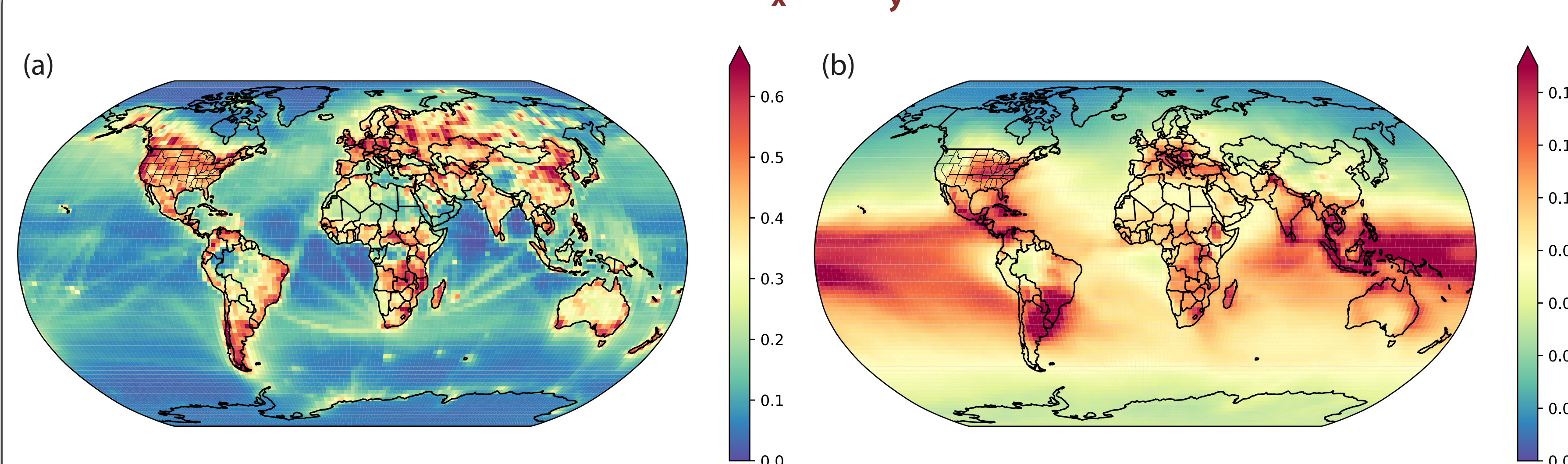


Figure 3: Modeled $\text{NO}_x : \text{NO}_y$ at (a) the surface and (b) 500 hPa.

- High $\text{NO}_x : \text{NO}_y$ regions are indicative of fresh NO_x emissions near the surface, with peaks near urban centers, major shipping lanes, and biomass burning regions.
- In the free troposphere, much of the available NO_x is consumed to form longer-lived NO_z , and lower $\text{NO}_x : \text{NO}_y$ ratios reflect mixing and transport.